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Data Visualization Report using Matplotlib and Seaborn

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1. Introduction

Data visualization is a core step in data analysis because it supports understanding patterns, trends, and relationships that may be difficult to detect from raw tables alone. The objective of this work is to apply visualization principles using Matplotlib and Seaborn to analyze three commonly used datasets—Iris, Tips, and Flights—and to communicate insights through appropriate chart selection and clear visual design. Emphasis is placed on both producing meaningful plots and interpreting them in a concise, evidence-driven manner.

2. Materials and Methods

2.1. Datasets

Three datasets were analyzed:

- **Iris dataset:** 150 observations of flower measurements (four numeric features) across three species.
- **Tips dataset:** 244 observations of restaurant transactions including bills, tips, and categorical attributes (e.g., time, day, smoker).
- **Flights dataset**: 144 monthly airline passenger counts spanning 1949–1960.

2.2. Exploratory Data Analysis (EDA)

To ensure basic data integrity and to understand variable types and ranges, a short EDA was conducted for each dataset, including:

- dataset shape (rows and columns),
- data types and structure (info()),
- summary statistics (describe()),
- and missing value checks (isnull().sum()).

For the Flights dataset, a time-series-ready index was created by constructing a date variable from year and month and sorting chronologically.

2.3. Visualization Approach

Visualizations were organized to cover key visualization categories:

- **Relationship plots**: scatter plots for numeric-to-numeric relationships.
- **Time-series plots:** line plots to show temporal trends.
- **Distribution plots:** histograms with KDE overlays and ECDFs to assess spread and skew.
- **Categorical comparisons:** bar, box, and violin plots to compare groups.
- **Heatmaps and dashboards:** correlation heatmaps and a 2×2 subplot grid (Matplotlib OOP) to present multiple views in one layout.

Both Seaborn and Matplotlib were used to demonstrate differences in workflow and control. Styling choices (themes, palettes, grid settings, and font/line parameters) were applied to improve readability and professionalism.

3. Results and Discussion

3.1. Iris Dataset

Scatter and multivariate plots were used to examine separability among species and relationships among features. The petal-based scatter plots indicate that setosa forms a distinct cluster, suggesting clear separability from the other species in this feature space. In contrast, versicolor and virginica overlap, implying that classification between these two species may require additional features or more complex decision boundaries.

The correlation heatmap supports the relationship observations: petal length and petal width exhibit strong positive association, indicating that these measurements increase together. Distribution plots (e.g., sepal length by species) show systematic differences in central tendency across species, reinforcing that the measured traits contain meaningful biological variation.

3.2. Tips Dataset

The Tips dataset was analyzed using scatter plots, distribution plots, and categorical comparison charts to understand tipping behavior. The scatter plot of total bill versus tip shows a general positive relationship, suggesting that larger bills are associated with higher tips; however, the spread at similar bill values indicates substantial variability in tipping practices.

Distribution plots show that tips are right-skewed, with most values concentrated at lower amounts and a smaller number of larger tips forming a long tail. Categorical plots (bar/box/violin) provide complementary perspectives: bar plots summarize group means, while box and violin plots highlight median behavior, variability, and outliers. Together, these views support comparisons across categories such as day and smoker status and emphasize that group differences should be interpreted with both central tendency and dispersion in mind.

3.3. Flights Dataset

Time-series and heatmap visualizations were used to analyze long-term change and seasonality. The line plot demonstrates a clear upward trend in passenger counts from 1949 to 1960, indicating substantial growth in air travel demand during the period. The month-by-year pivot heatmap reveals a consistent seasonal structure, where passenger volumes peak during certain months (commonly in mid-year) and decrease during other periods.

Importantly, the data exhibits both seasonality and growth: the seasonal cycle repeats annually, but peak values rise over time. This combined pattern is characteristic of many real-world time series where long-term changes coexist with periodic behavior.

4. Brief Note on Plotly

Plotly is a visualization library designed primarily for interactive charts. Unlike static figures, Plotly outputs typically allow zooming, panning, hovering for exact values, and interactive filtering. This makes Plotly especially useful for exploratory analysis and for sharing results through web-based dashboards (often via Dash). In contexts where user interaction and inspection of individual points are important, Plotly is a strong option; for static reporting and publication-style figures, Matplotlib/Seaborn are often preferred due to simplicity and formatting control.

5. Matplotlib vs Seaborn: Differences and Use Cases

5.1. Key Differences

- Abstraction level:
  - Matplotlib offers low-level, flexible control over every component (axes, ticks, layout, custom elements).
  - Seaborn provides higher-level functions optimized for statistical visualization and grouped comparisons.
- Default aesthetics:
  - Seaborn generally produces visually appealing plots with minimal configuration due to themes and palettes.
  - Matplotlib often requires more explicit styling to achieve the same level of visual polish.
- Statistical and categorical plotting:
  - Seaborn excels at box/violin/bar plots and hue-based comparisons with concise code.
  - Matplotlib can replicate these but often with more manual setup.

5.2. When to Use Each

- Use Seaborn when rapid development of clean statistical plots is needed, particularly for comparisons across categories (e.g., hue, distributions, summary plots).
- Use Matplotlib when fine control over layout, subplots, and publication formatting is required, or when building multi-panel dashboards with precise customization.

A practical workflow is to use Seaborn for plotting and Matplotlib for final figure refinement (layout, annotations if desired, and consistent formatting).

6. Conclusion

This work demonstrated the use of Matplotlib and Seaborn to visualize and interpret patterns across three datasets. The Iris dataset showed strong species separability for setosa and strong associations between petal features. The Tips dataset revealed a positive but variable bill–tip relationship and a right-skewed distribution of tips, with meaningful group comparisons supported by categorical plots. The Flights dataset exhibited pronounced long-term growth alongside stable seasonal patterns. Overall, Seaborn enabled efficient statistical visualization, while Matplotlib supported greater control over presentation and figure organization.